

PERSONALIZED MENTAL HEALTH DIGITAL TWIN USING AI

PROJECT PROPOSAL – IITISoC

PROBLEM DOMAIN: AI/ML

DIFFICULTY: ADVANCED

Team Details

Team Leader

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Team Member 3

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Team Member 4

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Team Member 5

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Skills: C++, Python

Domain: AI/ML

PS Name: Personalized Mental Health Digital Twin using AI

PS Number: AI/ML-303

Preference Number: 1

Project Summary

Mental health conditions often develop gradually and differ significantly across individuals, making early identification difficult. This project aims to develop a Personalized Mental Health Digital Twin capable of learning behavioural and emotional patterns from user activity, establishing personalized baselines, detecting deviations, and providing interpretable insights into possible mental health risks.

Project Solution

1. Journal and Behaviour Tracking

The system will allow users to maintain journal entries and track behavioural indicators such as sleep and mood patterns, and journaling consistency. This collected data will be used for analysing emotional and behavioural trends over time.

2. Sentiment and Behaviour Analysis

User entries will be analysed using sentiment analysis and behavioural trend analysis techniques. The system will attempt to identify patterns in emotional expression, consistency, and behavioural variations from the collected data.

User data is converted into measurable values. For example:

Behaviour	Possible Feature
Emotional expression	Sentiment score
Text pattern	Average sentence length and time
Sleep pattern	Sleep variance
Activity consistency	Journaling frequency

3. Personalized Behavioural Baselines

The system will create personalized behavioural baselines for individual users instead of relying on universal metrics. These baselines will help the system understand what is considered normal behavioural activity for a particular user.

For example:

Instead of saying: “*Anyone with a sentiment score < -0.5 is at risk.*”

We ask: “*What’s **normal** for **this** user?*”

User	Normal Sentiment Score
A	+0.6
B	-0.1

Now if User A’s score becomes -0.3, that’s a very major deviation whereas for User B that same score might not be unusual.

4. Anomaly Detection and Risk Assessment

Unusual behavioural or emotional deviations will be identified using anomaly detection techniques. Based on observed trends and deviations, the system will assess the risks of shifting towards negative mental states (either chronic or acute).

5. Dashboard, Visualizations, and Explainability

The frontend will include an interactive dashboard displaying graphs related to sleep patterns, sentiment variations, and behavioural changes over time. A major focus will be placed on **explainability**, privacy, and modular system design to ensure understandable outputs and easier future improvements.

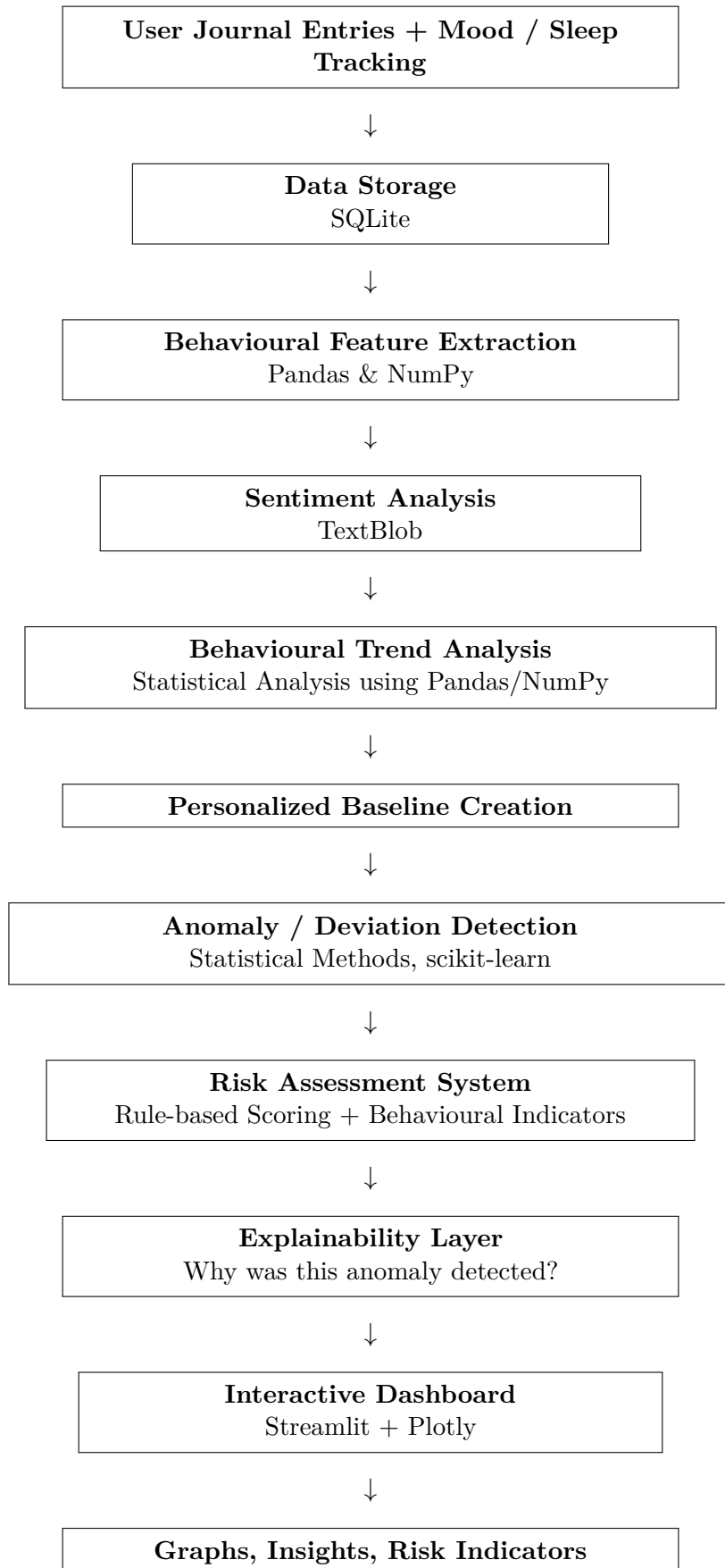
While giving the output, the system could ask:

“*What factors caused an anomaly in the behaviour to be detected?*”

An example output would then be:

- Increased negative sentiment
- Reduced journaling consistency
- Significant sleep irregularity

Proposed System Workflow



Key Deliverables

- **GitHub Repository:** A clean and reproducible GitHub repository containing the complete source code, documentation, and modular implementation of the system.
- **Technical Report:** A final report documenting the problem formulation, system architecture, implementation methodology, behavioural analysis techniques, observations, and limitations of the project.
- **Interactive Dashboard:** A frontend dashboard displaying behavioural trends, sleep patterns, sentiment variations, anomaly indicators, and explainable insights through interactive visualizations.
- **Demonstration Prototype:** A working prototype demonstrating journaling, behavioural monitoring, sentiment analysis, anomaly detection, and dashboard-based explainability features in real time.

Project Timeline

- **Week 1:** Familiarization with Python, AIML concepts, and overall project planning.
- **Week 2:** Development of journaling, mood tracking, sleep tracking, and data storage systems.
- **Week 3:** Implementation of sentiment analysis and behavioural trend analysis.
- **Week 4:** Development and testing of personalized baseline creation and anomaly/deviation detection systems.
- **Week 5:** Improvement and refinement of behavioural analysis, anomaly detection, and risk assessment systems.
- **Week 6:** Development of graphs, dashboard, visualizations, and explainability features.
- **Week 7:** Testing, debugging, optimization, and UI improvements.
- **Week 8:** Final polishing, documentation, mentor feedback integration, deployment, and presentation preparation.

Tech Stacks

These are some possible tools that we would like to use for this project, a skeleton per se. Please note that these tools are heavily subject to change as we improve our understanding of the architecture of this project and start optimising our system, though we will keep the framework the same.

Purpose	Tool
Language	Python
Dashboard/UI	Streamlit
Data Storage	SQLite
Data Analysis	Pandas & NumPy
Sentiment Analysis	TextBlob
Anomaly Detection	Statistical methods, scikit-learn
Graphs	Plotly
Version Control	GitHub

References

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4. GeeksforGeeks-Sentiment Analysis using Python
<https://www.geeksforgeeks.org/python-sentiment-analysis-using-vader/>
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